

Lab 8: Flask Fundamentals

Course: CST150: Web technologies

Objective

By the end of this lab, students will:

- Understand the basics of Flask, a Python web framework.
- Set up a simple Flask web application.
- Handle routing and render templates using Flask.

As always, don't just follow the workshop verbatim, tweak the changes and see the difference. Share your findings with the group.

Pre-requisites

- Completion of Lab 6 (Frontend Page Layout and Input Processing).
 - Python installed
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Lab Tasks

Task 1: Setting Up Flask

1. Create a `CST150_lab8` folder.
2. Open a terminal and create a virtual environment:

```
python -m venv flask_env
source flask_env/bin/activate

Or

flask_env\Scripts\activate

python -m flask --version
```

2. Install Flask:

```
pip install flask
```

3. Create a new file named `app.py` and add the following code:

```
from flask import Flask
app = Flask(__name__)
@app.route('/')
def home():
    return "Hello, Flask!"
if __name__ == '__main__':
    app.run(debug=True)
```

4. Run the Flask app:

```
python app.py
```

5. Open a browser and go to <http://127.0.0.1:5000/> to see your Flask app running.

Task 2: Handling Routes

1. Modify `app.py` to include multiple routes:

```
@app.route('/about')
def about():
    return "This is the About Page"
```

2. Restart the Flask app and test <http://127.0.0.1:5000/about>.

Task 3: Rendering HTML Templates

1. Create a `templates` folder in the project directory.
2. Inside `templates`, create a file named `index.html`:

```
<!DOCTYPE html>
<html>
<head>
    <title>Flask App</title>
</head>
<body>
    <h1>Welcome to Flask!</h1>
</body>
</html>
```

3. Modify `app.py` to render the template:

```
from flask import render_template

@app.route('/')
def home():
    return render_template('index.html')
```

4. Restart the Flask app and check `http://127.0.0.1:5000/` to see the rendered template.

Discussion Questions

1. What are the benefits of using Flask for web development?
2. How does Flask's routing mechanism work?
3. How can you pass dynamic data to an HTML template in Flask?

Bonus Challenge

- Add a new route `/contact` and create a `contact.html` page.
- Use Flask's `url_for` function to dynamically link pages.

Conclusion

In this lab, students set up a basic Flask web application, learned about routing, and rendered templates. Future lessons will introduce handling forms and integrating with a database.

End of Lab 8